LESSON 2.4

REWRITING SUMS USING THE DISTRIBUTIVE PROPERTY

LESSON INTRODUCTION

MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.

LEARNING TARGETS

- I can rewrite the sum of two composite numbers with a common factor using the distributive property.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

ESSENTIAL QUESTIONS

- How can the distributive property be used to generate equivalent expressions?
- How are common factors used to generate equivalent expressions for a sum of two composite numbers?

LESSON NARRATIVE

Students build on previous work with common factors of numbers in this unit and revisit using the distributive property to rewrite the sum of two composite numbers with a common factor. Students collaborate and discuss strategies to find missing values to make equations, comparing equivalent numeric expressions and analyzing errors when rewriting expressions using the distributive property. The Guided Instruction and Collaboration in this lesson provide opportunities to encourage flexible math thinking by rewriting equivalent expressions in various ways, including with common factors and the GCF.

COMPONENT SUMMARY

2.4.1 WARM-UP *(~5 MIN.)*

Students use clues to identify a “secret number”

2.4.2 COLLABORATION *(5-10 MIN.)*

Compare equivalent expressions and explain reasoning with partners

2.4.3 GUIDED INSTRUCTION *(10-20 MIN.)*

Analyze thinking and errors when generating equivalent expressions

2.4.4 COLLABORATION *(5-15 MIN.)*

Work with a partner to find missing values in equations

2.4.5 YOUR TURN! *(5-10 MIN.)*

Generate multiple equivalent expressions using common factors and the GCF

2.4.6 WRAP-UP *(~ 5 MIN.)*

Demonstrate understanding of learning targets by rewriting equivalent expressions using the distributive property

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Distributive property of multiplication over addition
- Greatest common factor (GCF)

MATERIALS

- (Optional) Chart paper
- (Optional) Sticky notes
- (Optional) Markers

ADDITIONAL PREPARATION

- If using a shared space for students to contribute ideas during 2.4.5 Your Turn!, prepare chart paper or a whiteboard space with the expressions from questions 1a-d. Gather sticky notes or markers for students to write generated equivalent expressions.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students who struggle with multiplication may struggle to identify factors and equivalent expressions.
- Students may confuse factors and multiples.

THINGS TO CONSIDER

- Consider using material and video from Lesson 1 and Lesson 3 to work on common factors.
- While Guided Instruction and Your Turn include prompts to rewrite expressions using the GCF, not every expression needs to be rewritten using the GCF. Encourage flexible thinking to rewrite expressions using both the GCF and other common factors.

LITERACY AND LANGUAGE ROUTINES

Clarify, Critique, and Correct

2.4.3 Guided Instruction provides an opportunity for students to develop math reasoning by analyzing errors when generating equivalent expressions. Consider prompting students to identify incorrect use of common factors or the distributive property, then correcting errors.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

Guided Instruction 2.4.3 positions students to analyze the math thinking of others and lets students recognize errors to suggest how to correctly rewrite an equivalent expression. Consider using these opportunities to engage in discussions about mathematical ideas effectively using math vocabulary.

DIFFERENTIATE INSTRUCTION

2.4.4 Collaboration offers a variety of entry points for different learning styles and can offer a variety of ways for students to engage with the activity. Use informal observation to determine which method best fits your classroom.

2.4.5 Your Turn! - students can post generated expressions in a shared space (e.g., class whiteboard or chart paper) using sticky notes or markers. Use this alternative process to check understanding and engage in discussion on identifying equivalent expressions since some questions have more than one correct answer.

For students who finish 2.4.5 or 2.4.6 early, consider extending the Secret Number activity as a challenge to complete with a friend or family member at home.

2.4.1 WARM-UP: SECRET NUMBER

Component Overview: *Students use a series of clues related to number properties to identify a secret number. Consider having students think of their own number, then use similar math vocabulary to create clues for somebody else to identify their number.*

QUESTIONING

Looking for an engaging idea for a project? Consider extending this activity by allowing students to write a series of interesting clues using factors, multiples, and other properties needed for somebody else to identify their secret number. This could also be an engaging homework activity.

2.4.1 Warm-Up: Secret Number

Juanita has a secret number. Read her clues and answer the questions that follow.

Clue 1: My secret number is a factor of 60.

1. **What is the smallest value Juanita's number could be?**
   1
2. **What is the largest value Juanita's number could be?**
   60

Clue 2: My number is prime.

3. **What values are possible for Juanita's number at this point?**
   2, 3, or 5

Clue 3: 15 is a multiple of my secret number.
3 or 5

Clue 4: My secret number is a factor of 20.

4. **What is Juanita's secret number?**
   5
5. **Could you have determined her secret number only using the first three clues? Explain.**
   No, it only narrowed the answer down to 2 possibilities.

2.4.2 Collaboration: Equivalent Numeric Expressions

In each equation, two numeric expressions are set equal to each other.

1. With your partner, discuss whether each equation is true or false and complete the table.

Equation	Is it true or false?
$42 + 12 = 6(7 + 2)$	*True*
$6(7 + 2) = (6 \cdot 7) + (6 \cdot 2)$	*True*

2. How is 6 related to 42 and 12?
6 is a common factor of both 42 and 12.

3. How do the values 7 and 2 relate to the original values 42 and 12?
They are the factors multiplied by 6 to get 42 and 12.

2.4.2 COLLABORATION: EQUIVALENT NUMERIC EXPRESSIONS

Component Overview: *Students work in partners to identify if expressions are equivalent or not. Consider giving students an opportunity to verbalize and reason about strategies to determine if the expressions are equivalent, focusing on recognizing the relationship of common factors and properties of operations.*

TEACHER MOVES

Teachers should observe how students determine whether the two expressions are equivalent or not. Some will justify using the distributive property, while others may actually evaluate both sides of each equation. Connections should be made between both ways of thinking.

TEACHER MOVES

Use this collaboration to encourage students to reason and verbalize strategies for determining if expressions are equivalent. There is more practice to rewrite expressions in the next activities. Provide sentence stems to foster discussion, including “I think the equation is true because…” or “My strategy to determine if the equation is true is…”

ENRICHMENT

What makes an equation true? Both sides of the equation must be equivalent to each other.

2.4.3 GUIDED INSTRUCTION: REWRITING SUMS USING DISTRIBUTIVE PROPERTY

Component Overview: *Students rewrite the sum of 16 and 24 using common factors and the distributive property, then analyze errors in rewritten expressions that are not equivalent. This activity provides a further opportunity for group or partner discussion to recognize, correct, and clarify procedural use of the distributive property to rewrite equivalent expressions.*

ENRICHMENT

Consider activating prior knowledge from previous lessons:

- What is a factor? What is a common factor?
- Why would we need to be looking for common factors?
- Which factors can be used to create equivalent expressions?

DIFFERENTIATED INSTRUCTION

Notice the icons used to represent different parts of the working process. Consider using these "thinking" and "work" prompts for your students to create additional opportunities for them to make sense of information.

QUESTIONING

The error analysis embedded within this activity gives students additional practice on MA.K12.MTR.4.1 and higher order thinking.

TEACHER MOVES

Clarify, Critique, and Correct

Consider emphasizing and asking students to identify and share reasons for why the errors lead to expressions that are not equivalent. Focus on reasoning connected to the correct use of the distributive property and identification of common factors.

2.4.3 Guided Instruction: Rewriting Sums Using Distributive Property

Sums of ***composite whole numbers*** can be rewritten using the distributive property and a common factor of the numbers.

A ***composite number*** is a whole number greater than 1 that has at least one whole-number factor other than one and itself.

1. What are the common factors of 16 and 24?

Factors of 16	1, 2, 4, 8, 16
Factors of 24	1, 2, 3, 4, 6, 8, 12, 24
Common Factors of 16 and 24	1, 2, 4, 8

2. Isaiah rewrote the sum 16 + 24 using the distributive property. Each step in his thinking is shown. Complete the descriptions of Isaiah's process by examining his thinking and work. Use the words in the bank below to complete each description.

distributive	equivalent	factor	multiplication

Isaiah's Thinking and Work	Description
"I know 2 is a common factor of both 16 and 24."	Isaiah identified a *factor* both numbers have in common.
$16 + 24 = 2(8) + 2(12)$	He rewrote the expression using *multiplication* with the common factor.
$2(8) + 2(12) = 2(8 + 12)$	He rewrote the expression using the *distributive* property.
"The sum $16 + 24$ can be rewritten as $2(8 + 12)$ using the distributive property."	Isaiah recognized that $16 + 24$ and $2(8 + 12)$ are *equivalent* numeric expressions.

3. Rewrite the sum 16 + 24 using the distributive property in two additional ways.
   $16 + 24 = 8(2 + 3)$

   $16 + 24 = 4(4 + 6)$

   $2(8 + 12)$ is also a possible solution.

4. Two students made errors rewriting the sum 54 + 36 using the distributive property. Each student's expression is shown.

Rewrite 54 + 36 using the distributive property.	
Naveah's Expression	4(14 + 9)
Gavin's Expression	3(18 + 36)

Work with your partner to answer the following questions.

a. What error did Naveah make?
   $4 \cdot 14 \neq 54$
   4 is not a common factor of 54 and 36.

b. What error did Gavin make?
   Gavin forgot to factor the 3 out of the 36.

c. Generate two different equivalent expressions for the sum 54 + 36 by rewriting it using the distributive property.
   Answers will vary.
   $2(27 + 18)$
   $3(18 + 12)$

5. Rewrite an equivalent expression for the sum of 40 + 64 using the distributive property with the **greatest common factor** (GCF).
   $8(5 + 8)$

2.4.4 Collaboration: Missing Values

1. Work with your partner to determine the missing values that will make each equation true.

   a. $40 + 16 = 4(10 + 4)$
   b. $15 + 80 = 5(3 + 16)$
   c. $108 + 198 = 18(6 + 11)$
   d. $28 + 63 = 7(4 + 9)$
   e. $45 + 18 = 3(15 + 6)$
   f. $135 + 27 = 9(15 + 3)$

2. Choose one of the equations from question 1 and explain how you determined the missing value(s).
   Answers will vary.

   To find the missing value in equation __________, I ____

   ____.

3. Have your partner read your explanation and offer suggestions on how to improve it.

2.4.4 COLLABORATION: MISSING VALUES

Component Overview: *Students work in partnerships to write missing values that make an equation true and write about how they determined the missing value. Consider having students verbalize reasoning for the values they used to make the equations true.*

TEACHER MOVES

Determine how students are collaborating based on informal observations from the first few activities.

- Have students split the work (1a-c and 1d-f) and share their answers and reasoning.
- Have students work together, with one student only allowed to talk (progressing student) and other taking notes (proficient student).
- Have students choose which three to engage with (increasing choice and entry point) and find a partner around the room who did the same three to compare answers for questions 2 and 3.

DIFFERENTIATED INSTRUCTION

Notice additional entry points using sentence stems in question 2 and intentional sharing for auditory learners in question 3. Consider using vocabulary cards to encourage students to replace words like “same number” with “common factor” when articulating their responses.

2.4.5 YOUR TURN!

Component Overview: *Students work independently to rewrite expressions using common factors, including the GCF and the distributive property. Consider using this practice to encourage collaboration, flexible thinking, and discussion by allowing students to post generated equivalent expressions in a shared space.*

QUESTIONING

Challenge students who finish early to create as many equivalent expressions as possible. They will notice that 1a and 1d only have one common factor, while option 1c has a variety.

2.4.5 Your Turn!

1. **Rewrite an equivalent expression for each sum using the distributive property with a common factor.**

   a. $51 + 21$
   $3(17 + 7)$

   b. $124 + 20$
   $2(62 + 10)$ *or* $4(31 + 5)$

   c. $72 + 24$
   $24(3 + 1)$
   $12(6 + 4)$
   $8(9 + 3)$
   $6(12 + 4)$
   $4(18 + 6)$
   $3(24 + 8)$
   $2(36 + 12)$

   d. $39 + 65$
   $13(3 + 5)$

2. **Rewrite an equivalent expression for the sum of $68 + 34$ using the distributive property with the greatest common factor (GCF).**
   $34(2 + 1)$

2.4.6 Wrap-Up: Ticket Out the Door

1. Rewrite an equivalent expression for the sum of $75 + 30$ in two different ways using the distributive property and a common factor.
   Answers will vary.
   Possible answers include:
   $15(5+2)$ *or* $5(15+6)$ *or* $3(25+10)$

2. Complete the equation with the missing values that make the equation true.

$$49 + 35 = 7\,(7 + 5)$$

2.4.6 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Students apply their understanding to rewrite sums of two composite numbers using common factors and the distributive property.*

DIFFERENTIATED INSTRUCTION

Consider supplementing these questions by asking students to answer one of the essential questions for this lesson in their own words. The essential questions are open-ended and allow students to represent their thinking in a format that best communicates their understanding. For example, one student could create a visual, step-by-step guide with labels explaining their response.

